

Figura 1. Diseño preliminar del puente.

Fuente: Elaboración propia.

Longitud de las diagonales:

$$Ld = \sqrt{151^2 + 47^2} = 158,1455cm$$

$$\text{sen}\,\theta = \frac{151}{158,1455} = 0,9548 = S$$

$$\cos\theta = \frac{47}{158,1455} = 0,2972 = C$$

Reacciones:

$$\sum M_{A0} = 0$$

$$A_{7y}(658) - \frac{P}{8}(188) - \frac{P}{8}(282) - \frac{P}{8}(376) - \frac{P}{8}(470) = 0$$

$$658A_{7y} = \frac{1316P}{8}$$

$$A_{7y} = \frac{1316P}{5264}$$

$$A_{7y} = \frac{P}{4}$$

$$\sum F_y = 0$$

$$A_{0y} + A_{7y} - 4\left(\frac{P}{8}\right) = 0$$

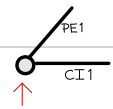
$$A_{0y} + \frac{P}{4} - \frac{P}{2} = 0$$

$$A_{0y} = \frac{P}{4}$$

$$\sum F_x = 0$$

$$A_{0x} = 0$$

Nodo A_0



$\sum F_y = 0$

$\frac{P}{4} + F_{PE1}S = 0$

$F_{PE1} = -\frac{P}{4S}$

$F_{PE1} = -0,2618P$ (compresión)

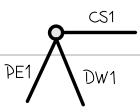
$\sum F_x = 0$

$F_{CI1} + F_{PE1}C = 0$

$F_{CI1} = -(-0,2618P)C$

$F_{CI1} = 0,0778P$ (tensión)

Nodo U1



$\sum F_y = 0$

$-F_{PE1}S - F_{DW1}S = 0$

$-F_{PE1} - F_{DW1} = 0$

$F_{DW1} = -F_{PE1}$

$F_{DW1} = 0,2618P$ (tensión)

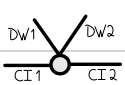
$\sum F_x = 0$

$-F_{PE1}C + F_{DW1}C + F_{CS1} = 0$

$F_{CS1} = F_{PE1}C - F_{DW1}C$

$F_{CS1} = -0,1556P$ (compresión)

Nodo A1



$\sum F_y = 0$

$F_{DW1}S + F_{DW2}S = 0$

$F_{DW2}S = -F_{DW1}S$

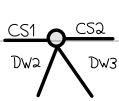
$F_{DW2}S = -0,2618P$ (compresión)

$\sum F_x = 0$

$-F_{CI1} - F_{DW1}C + F_{DW2}C + F_{CI2} = 0$

$F_{CI2} = 0,2334P$ (tensión)

Nodo U2



$\sum F_y = 0$

$-F_{DW2}S - F_{DW3}S = 0$

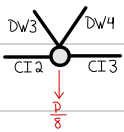
$F_{DW3} = 0,2618P$ (tensión)

$\sum F_x = 0$

$-F_{DW2}C + F_{DW3}S - F_{CS1} + F_{CS2} = 0$

$F_{CS2} = -0,3113P$ (compresión)

Nodo A1



$\sum F_y = 0$

$F_{DW3}S + F_{DW4}S - \frac{P}{8} = 0$

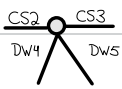
$F_{DW4} = -0,1309\,P$ (compresión)

$\sum F_x = 0$

$-F_{CI2} - F_{DW3}C + F_{DW4}C + F_{CI3} = 0$

$F_{CI3} = 0,3502P$ (tensión)

Nodo U3



$\sum F_y = 0$

$-F_{DW4}S - F_{DW5}S = 0$

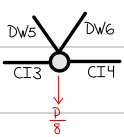
$F_{DW5} = 0,1309P$ (tensión)

$\sum F_x = 0$

$-F_{DW4}C + F_{DW5}C - F_{CS2} + F_{CS3} = 0$

$F_{CS3} = -0,3891P$ (compresión)

Nodo A3



$\sum F_y = 0$

$F_{DW5}S + F_{DW6}S - \frac{P}{8} = 0$

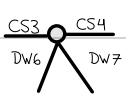
$F_{DW6} = 0$

$\sum F_x = 0$

$-F_{CI3} - F_{DW5}C + F_{DW6}C + F_{CI4} = 0$

$F_{CI4} = 0,3891P$ (tensión)

Nodo U4



$\sum F_y = 0$

$-F_{DW6}S - F_{DW7}S = 0$

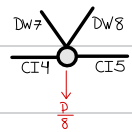
$F_{DW7} = 0$

$\sum F_x = 0$

$-F_{CS3} + F_{CS4} - F_{DW6}C + F_{DW7}C = 0$

$F_{CS4} = -0,3891P$ (compresión)

Nodo A3



$\sum F_y = 0$

$F_{DW7}S + F_{DW8}S - \frac{P}{8} = 0$

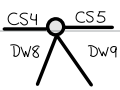
$F_{DW8} = 0,1309P$ (tensión)

$\sum F_x = 0$

$-F_{CI5} - F_{DW7}C + F_{DW8}C + F_{CI5} = 0$

$F_{CI5} = 0,3502P$ (tensión)

Nodo U4



$\sum F_y = 0$

$-F_{DW8}S - F_{DW9}S = 0$

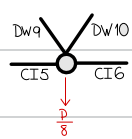
$F_{DW9} = -0,1309P$ (compresión)

$\sum F_x = 0$

$-F_{CS4} + F_{CS5} - F_{DW8}C + F_{DW9}C = 0$

$F_{CS5} = -0,3113P$ (compresión)

Nodo A5



$\sum F_y = 0$

$F_{DW9}S + F_{DW10}S - \frac{P}{8} = 0$

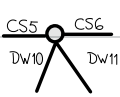
$F_{DW10} = 0,2618P$ (tensión)

$\sum F_x = 0$

$-F_{CI5} - F_{DW9}C + F_{DW10}C + F_{CI6} = 0$

$F_{CI6} = 0,2334P$ (tensión)

Nodo U6



$\sum F_y = 0$

$-F_{DW10}S - F_{DW11}S = 0$

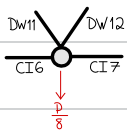
$F_{DW11} = -0,2618P$ (compresión)

$\sum F_x = 0$

$-F_{CS5} + F_{CS6} - F_{DW10}C + F_{DW11}C = 0$

$F_{CS6} = -0,1556P$ (compresión)

Nodo A6



$\sum F_y = 0$

$F_{DW11}S + F_{DW12}S = 0$

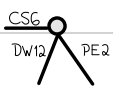
$F_{DW12} = 0,2618P$ (tensión)

$\sum F_x = 0$

$-F_{CI6} - F_{DW11}C + F_{DW12}C + F_{CI7} = 0$

$F_{CI7} = 0,0778P$ (tensión)

Nodo U7



$\sum F_y = 0$

$-F_{DW12}S - F_{PE2}S = 0$

$F_{PE2} = -0,2618P$ (compresión)

$\sum F_x = 0$

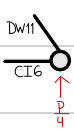
$-F_{CS6} - F_{DW12}C + F_{PE2}C = 0$

$F_{CS6} = -0,1556P$ (compresión)

$-(-0,1556P) - (0,2618P)(0,2972) + (-0,2618P)(0,2972) = 0$

$0,1556P - 0,0778P - 0,0778P \approx 0$

Nodo A7



$\sum F_y = 0$

$A_{7y} + F_{PE2}S = 0$

$\frac{P}{4} + (-0,2618P)(0,9548)$

$\sum F_y \approx 0$

$\sum F_x = 0$

$-F_{CI7} - F_{PE2}C = 0$

$-0.0778P - (-0,2618P)(0,2972)$

$\sum F_x \approx 0$